



BALE Top-of-Book Market Data Feed

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0.1 Revision History

Date	Version	Notes
March 03, 2026	0.1.draft	Initial version.
May 08, 2026	0.2.draft	Added listingMarket field to DefineSymbol.
May 10, 2026	0.3.draft	Added tradeThroughExemptionReason field to the Trade message.
May 15, 2026	0.4.draft	Added auction status messages: AuctionPreamble, AuctionBandWindow, and AuctionPrint.
June 08, 2026	0.5.draft	Documented how messages may be extended in future releases.
June 12, 2026	1.0	TXSE Specification for Trading.

1 Overview

This document details the complete specification for the TXSE (Texas Stock Exchange) Top-of-Book Market Data Feed. This feed is designed to provide real-time, best bid and offer (top-of-book) market data for all trading streams on the Texas Stock Exchange.

BALE does not include a framing or session protocol. These are handled by a higher-level protocol such as RAKE TCP or RAKE UDP for multicast distribution.

The specification covers symbol definitions (DefineSymbol), symbol status (SymbolStatus), trading session status (TradingSessionStatus), individual executions (Trade), the combined top-of-book message (TopOfBook), and the auction status messages (AuctionPreamble, AuctionBandWindow, and AuctionPrint).

Message ordering: when an execution causes both Trade and TopOfBook messages, all Trade messages from that execution event are delivered before the TopOfBook that reflects the resulting depth change.

1.1 Optional Fields / Presence bits

Presence bits allow messages to include only the optional fields actually needed. Each bit indicates whether an optional field is present, minimizing bandwidth usage while maintaining message flexibility.

Message Structure

When optional fields are present, messages follow this layout:

```
[Message Type] [Presence Bits] [Fixed Fields] [Optional Field 1] [Optional Field 2] ... [Optional Field N]
```

The presence bits appear immediately after the message type identifier and before the fixed fields. Optional fields are appended in bit order after all fixed fields.

Example: TradingSessionStatus with RegulatoryHaltReason

```
Message Type: 'i' (0x69)
Presence Bits: 0x02 (bit 1 set)
Fixed Fields: [transactTime] [marketHoursState] [tradingState]
Optional Fields: [regulatoryHaltReason (1 byte)]
```

In this example, presence bit 0x02 indicates that regulatoryHaltReason (bit 1) is included, while operationalHaltReason (bit 0) is omitted. Optional fields are always in the same order as their corresponding bit positions.

1.2 Message Extensibility

How the Exchange extends a message in a future release depends on whether that message carries optional fields via presence bits. In every case, Members should read the fields they recognize and ignore anything beyond them, rather than relying on a fixed message byte length.

Messages with only fixed fields

The Exchange may append new fields to the end of the message. Because each message is length-delimited by its framing, a Member running against an earlier specification reads the fields it recognizes and ignores the additional trailing bytes. For example, a future release could append a field to the end of DefineSymbol, which carries only fixed fields; an existing Member continues to read the fields it knows and ignores the rest.

Messages with optional fields via presence bits

The Exchange extends these by introducing additional optional fields, each assigned a presence bit that is currently unused; when present, a new optional field follows the existing optional fields. A Member running against an earlier specification should ignore any presence bit it does not recognize, along with the optional field that bit introduces. For example, `SymbolStatus` today uses presence bits 0 and 1 (`operationalHaltReason` and `regulatoryHaltReason`) and leaves bits 2 through 7 available; a future release could add an optional field on bit 2, which an existing Member would not recognize and would ignore.

These additions leave the size and position of every existing field unchanged and are not a breaking change. The Exchange will not insert a field among existing fields, reorder fields or presence bits, or widen a message's presence-bit field without a new protocol version. In particular, a presence-bit field with no unused bits remaining cannot accept further optional fields except under a new protocol version.

1.3 Auction Band Window and Late Limit Orders

Band Window Definitions

Opening Band Window: 9:28:00 ET until the Opening Auction at 9:30:00 ET. The following order types become non-cancelable at the start of this window: `AT_THE_OPEN` orders and `RHO` orders. The Participation Band published in the `AuctionBandWindow` message is dynamic and may shift on each 5-second dissemination cycle; the final `AuctionBandWindow` is published 5 seconds before the cross. The band used to adjust late-order prices is the band in effect at the auction print, which is not itself disseminated.

Closing Band Window: 3:58:00 ET until the Closing Auction at 4:00:00 ET. The following order types become non-cancelable at the start of this window: `AT_THE_CLOSE` orders. The Participation Band published in the `AuctionBandWindow` message is dynamic and may shift on each 5-second dissemination cycle; the final `AuctionBandWindow` is published 5 seconds before the cross. The band used to adjust late-order prices is the band in effect at the auction print, which is not itself disseminated.

Band Window Eligibility by TimeInForce

Band Window constraints (non-cancelability and Participation Band price adjustment) apply only to orders with auction-eligible TIFs:

- `AT_THE_OPEN` orders are auction-classed by TIF. They become non-cancelable at the start of the Opening Band Window (9:28:00 ET).
- `AT_THE_CLOSE` orders are auction-classed by TIF. They become non-cancelable at the start of the Closing Band Window (3:58:00 ET).
- `RHO` orders on TXSE-primary-listed symbols become non-cancelable at the start of the Opening Band Window (9:28:00 ET) and are subject to Opening Band Window price adjustment. After the Opening Auction prints, any unfilled shares enter the continuous book in entry-time order.

Orders entered with TIF `AT_THE_OPEN` during 9:28:00–9:30:00 ET are referred to as **LOO.L** (Late Limit-On-Open) and are subject to Participation Band price adjustment at the auction print. Orders entered with TIF `AT_THE_CLOSE` during 3:58:00–4:00:00 ET are referred to as **LOC.L** (Late Limit-On-Close) and are subject to Participation Band price adjustment at the auction print. Regular `AT_THE_OPEN` / `AT_THE_CLOSE` orders entered before the Band Window are auction-eligible but not price-adjusted; they participate at their entered limit price.

Orders with any other TIF (`DAY`, `SYS`, `IOC`, `GTT`) are not subject to the Band Window (no non-cancelability and no price adjustment). They route to the continuous book; if resting at a cross they participate in the auction at the clearing price as continuous-book liquidity (their volume does not appear in the auction-only band-edge share counts — see below).

Late Limit Auction Order Behavior with Participation Bands

During the Band Window, the share counts published every 5 seconds in `AuctionBandWindow` show eligible volume at each band edge. A late limit auction order with a limit price more aggressive than the Upper Band (for bids) or the Lower Band (for offers) is counted at the corresponding band edge in these published share counts. As the band is recalculated each cycle, an order's position relative to the band may change, and its contribution to the published share counts updates accordingly; this is evaluated fresh each cycle and is not carried forward — only the order's late designation persists.

At the auction print, the order's price is adjusted to be equal to the corresponding band edge — a buy above the Upper Band is adjusted to the Upper Band, a sell below the Lower Band is adjusted to the Lower Band — and the order may participate in the auction at the adjusted price.

Band-edge share counts are auction-only: they reflect `AT_THE_OPEN`, `AT_THE_CLOSE`, `LOO`, `LOC`, `LOO.L`, `LOC.L`, and `RHO` (during the Opening Band Window). Resting continuous-book orders (`DAY`, `SYS`, `IOC`, `GTT`, `peg`, `hidden`, and `reserve`) participate in the auction at the clearing price, but their volume does not appear in the band-edge share counts.

The excess auction side is not republished in the `AuctionBandWindow` message during the Band Window. The Exchange publishes the buy and sell shares eligible at each band edge; firms can derive the prevailing excess side from those share counts. The earlier `AuctionPreamble` cycle, which streams during the Accumulation Period, carries `excessAuctionSide` explicitly.

Worked examples (all share the same band, Lower \$99.80, Upper \$100.20):

1. **Limit inside the band.** A buy late limit auction order at \$100.10 participates at its entered limit; no price adjustment.
2. **Buy above the Upper Band.** A buy late limit auction order at \$101.00 has its price adjusted to the Upper Band (\$100.20) at the print and participates at that price. If the auction clears at \$100.05, the buy fills at \$100.05.
3. **Sell below the Lower Band.** A sell late limit auction order at \$99.50 has its price adjusted to the Lower Band (\$99.80) at the print and participates at that price. If the auction clears at \$100.10, the sell fills at \$100.10.

2 Types

- This document uses the terms Byte, Short, Int, Long to refer to numeric values that are 8,16,32,64 bit long respectively. All are two's complement signed little-endian encoded.

Type	Byte Length	MAX_VALUE
Byte	1	$2^7 - 1 = 127$
Short	2	$2^{15} - 1 = 32,767$
Int	4	$2^{31} - 1 = 2,147,483,647$
Long	8	$2^{63} - 1 = 9,223,372,036,854,775,807$

- Strings are all ASCII, fixed-length. Strings shorter than the fixed-length should be left justified, by right padded with spaces (ASCII 0x20) to the full string length
- Enums are encoded as a single Byte unless they appear in a bitfield, then they use the number of bits specified.
- TimeStamps are nanoseconds since the Unix epoch in UTC time zone, encapsulated by a Long
- Prices are a Long with an implied scale of 100,000,000. Each unit is \$0.00000001

2.1 AuctionType

Type of auction event.

Enum Constant	Value	Notes
OPENING	1	Scheduled regular-session cross that establishes the official opening price for a symbol. Cross fires at 9:30 AM ET. Includes an Accumulation Period (AuctionPreamble published, 8:00–9:28 AM ET) followed by a Band Window (AuctionBandWindow published, 9:28–9:30 AM ET).
CLOSING	2	Scheduled end-of-day cross that establishes the official closing price for a symbol. Runs once per trading day at the regular-session close. Includes both an Accumulation Period (AuctionPreamble published) and a Band Window (AuctionBandWindow published). Also used for halt-to-close transitions when a halt is unresolved at the end of the regular session (see RESUME_PENDING).
HALT	3	Reopening cross following a trading halt or LULD trading pause. Enters the Band Window directly at halt onset; no Accumulation Period and no AuctionPreamble are emitted. AuctionBandWindow is published every 5 seconds during the Band Window. May extend if the projected clearing price falls outside the Participation Band.
IPO	4	Cross for a symbol's initial public offering. Enters the Band Window directly at the IPO acceptance-window opening; no Accumulation Period and no AuctionPreamble are emitted. AuctionBandWindow is published every 5 seconds during the Band Window. Symbol tradingState is RESUME_PENDING during the Band Window prior to the cross.

2.2 AuctionExcessSide

Side with excess auction-book interest.

Enum Constant	Value	Notes
EQUAL	0	Buy and sell Auction-Only interest are balanced (equal volume on each side), or no Auction-Only interest is currently in the book.
BUY	1	Buy interest exceeds sell interest.
SELL	2	Sell interest exceeds buy interest.

2.3 MarketParticipantId

Market participant identifier for the primary listing market. Currently, this field is populated only when TXSE is the listing market; NOT_AVAILABLE (0x00) is used otherwise. Additional listing market values may be defined in future revisions to identify listing markets for other exchanges. Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.

Enum Constant	Value	Notes
NOT_AVAILABLE	0	TXSE is not the listing market for this symbol.
TXSE	70	Texas Stock Exchange, LLC. CTA CTS and UTP participant code 'F'.

2.4 SymbolTradingState

Symbol-specific trading state indicating current availability.

Enum Constant	Value	Notes
TRADING	1	Symbol is open and available for trading.
HALTED	2	Trading suspended. The cause, if available, is carried in the operationalHaltReason or regulatoryHaltReason optional fields of the accompanying Symbol-Status message.
RESUME_PENDING	3	Applies only when a symbol is in a non-trading state with an auction cross pending to bring it back into trading. Covers HALT reopening crosses and IPO crosses. The accompanying AuctionBandWindow message identifies the specific auction type.

2.5 ShortSaleRestrictionState

Short sale restriction state under SEC Rule 201.

Enum Constant	Value	Notes
NONE	0	No short sale restriction in effect (CTA: Space, UTP: 0).
ACTIVATED	1	Restriction activated intra-day (CTA: A, UTP: 1).
CONTINUED	2	Restriction carried over from previous trading day (CTA: C, UTP: 2).
DEACTIVATED	3	Restriction deactivated intra-day (CTA: D).
IN_EFFECT	4	Restriction in effect, status unchanged from previous close (CTA: E).

2.6 OperationalHaltReason

Operational and technical reasons for trading halts initiated by the Exchange. Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.

Enum Constant	Value	Notes
ADMINISTRATIVE	0	Administrative or technical halt initiated by the Exchange.

2.7 RegulatoryHaltReason

Regulatory reasons for trading halts and pauses. Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.

Enum Constant	Value	Notes
REASON_NOT_AVAILABLE	0	Reason not available or not applicable (CTA: Space, UTP: Space).
NEWS_PENDING	1	Trading halted pending release of material news (CTA: P, UTP: T1).
NEWS_DISSEMINATION	2	Trading halted for dissemination of material news (CTA: D, UTP: T2).
INFO_REQUESTED	3	Additional information requested by the listing exchange (CTA: A, UTP: T12).
ORDER_IMBALANCE	4	Order imbalance (CTA: I).
EXTRAORDINARY_MARKET_ACTIVITY	5	Extraordinary market activity (UTP: T6).
LULD_PAUSE	6	Limit Up-Limit Down pause (CTA: M, UTP: LUDP).
NON_COMPLIANCE	7	Non-compliance with listing standards (UTP: H4).
FILINGS_NOT_CURRENT	8	Required filings not current (UTP: H9).
SEC_SUSPENSION	9	SEC trading suspension (UTP: H10).
REGULATORY_CONCERN	10	Regulatory concern from other markets (CTA: C, UTP: H11).
SUB_PENNY_TRADING	11	Sub-penny trading violation (CTA: Y).
ETF	12	ETF-specific halt (CTA: F, UTP: T8).
IPO_NOT_TRADING	13	IPO not yet trading (UTP: IPO1).
CORPORATE_ACTION	14	Corporate action in progress (CTA: N, UTP: M1).
CIRCUIT_BREAKER_L1	15	Market-wide circuit breaker Level 1 (CTA: 1, UTP: MWC1).
CIRCUIT_BREAKER_L2	16	Market-wide circuit breaker Level 2 (CTA: 2, UTP: MWC2).
CIRCUIT_BREAKER_L3	17	Market-wide circuit breaker Level 3 (CTA: 3, UTP: MWC3).

Enum Constant	Value	Notes
CIRCUIT_BREAKER_CARRYOVER	18	Market-wide circuit breaker halt carried over from previous trading day (UTP: MWC0).
MERGER_EFFECTIVE	19	Merger effective (CTA: E).
NEW_SECURITY_OFFERING	20	New security offering (CTA: O).
INDICATIVE_VALUE_NOT_AVAILABLE	21	Intraday indicative value not available (CTA: V).
QUOTATION_NOT_AVAILABLE	22	Quotation not available (UTP: M2).
SINGLE_STOCK_TRADING_PAUSE	23	Single stock trading pause (UTP: T5).
OPERATIONAL	24	Non-regulatory halt due to technical or operational reasons (CTA: X, UTP: O1).
LULD_PAUSE_STRADDLE	25	Limit Up-Limit Down volatility pause - straddle condition (bid/offer crosses a band) (UTP: LUDS).
NEWS_RESUMPTION_TIMES	26	News and resumption times announced (UTP: T3).
SINGLE_STOCK_TRADING_PAUSE_QUOTING	27	Single stock trading pause (UTP: T7).
QUALIFICATIONS_ISSUES_RESOLVED	28	Qualifications issues reviewed/resolved; quotations/trading to resume (UTP: R4).
FILING_REQUIREMENTS_SATISFIED	29	Filing requirements satisfied/resolved; quotations/trading to resume (UTP: R9).
ISSUER_NEWS_NOT_FORTHCOMING	30	Issuer news not forthcoming; quotations/trading to resume (UTP: C3).
QUALIFICATIONS_HALT_ENDED	31	Qualifications halt ended; maintenance requirements met; resume (UTP: C4).
QUALIFICATIONS_HALT_CONCLUDED	32	Qualifications halt concluded; filings met; quotes/trades to resume (UTP: C9).
REGULATORY_AUTHORITY_HALT_CONCLUDED	33	Trade halt concluded by other regulatory authority; quotes/trades to resume (UTP: C11).
NEW_ISSUE_AVAILABLE	34	New issue available (UTP: R1).
ISSUE_AVAILABLE	35	Issue available (UTP: R2).
IPO_QUOTATION_RELEASED	36	IPO security released for quotation (UTP: IPOQ).
IPO_POSITIONING_WINDOW_EXTENSION	37	IPO security - positioning window extension (UTP: IPOE).

Enum Constant	Value	Notes
CIRCUIT_BREAKER_QUOTATION_RESUMPTION	38	Market-wide circuit breaker quotation resumption (UTP: MWCQ).

2.8 TradeThroughExemptionReason

Sale condition byte 2: Whether this trade is exempt from Reg NMS Rule 611 trade-through protection

(CTA/UTP sale condition category/level 2). Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.

Enum Constant	Value	Notes
NONE	0	No trade-through exemption.
INTERMARKET_SWEEP	1	Intermarket Sweep Order (ISO) (CTA/UTP: 'F').

2.9 MarketHoursState

Current phase of the market trading day.

Enum Constant	Value	Notes
CLOSED_BEFORE_HOURS	0	Markets are closed before trading hours.
EARLY_SESSION	1	Early trading session is active (pre-market).
REGULAR_SESSION	2	Regular trading session is active.
AFTER_HOURS_SESSION	3	After-hours trading session is active (post-market).
CLOSED_AFTER_HOURS	4	Markets are closed after trading hours.

2.10 SessionTradingState

Market-wide trading state indicating the default state for all symbols (individual symbols may have additional restrictions).

Enum Constant	Value	Notes
CLOSED	0	Market session is closed - outside of trading hours or not yet opened.
TRADING	1	Market session is open and trading is active (individual symbols may still be halted/paused).
HALTED	2	Market-wide trading halt - all symbols halted (circuit breaker or system-wide halt).

3 BALE Messages

3.1 TradingSessionStatus

Market-wide status representing the overall state of the trading session.

Field Name	Offset	Size	Type	Notes
messageType	0	1	Byte	'i'/0x69/105
presenceBits	1	1	Byte	See Optional Fields
transactTime	2	8	TimeStamp	
marketHoursState	10	1	Enum	Enum MarketHoursState . Current phase of the market trading day.
tradingState	11	1	Enum	Enum SessionTradingState . Market-wide trading state indicating the default state for all symbols (individual symbols may have additional restrictions).

3.1.1 TradingSessionStatus Presence Bits

These presence bits are as follows. Any added optional fields must be in their presence bit order [See Optional Fields](#)

Bit Name	Set Mask
hasOperationalHaltReason	0x01
hasRegulatoryHaltReason	0x02
reserved	0xFC

3.1.2 TradingSessionStatus Optional Fields

The optional fields begin directly after the end of the field above. If a bit is set in the presence bits, the field must be present, and in the order listed in the presence bits. [See Optional Fields](#)

Field Name	Size	Type	Notes
operationalHaltReason	1	Enum	Enum OperationalHaltReason . Operational and technical reasons for trading halts initiated by the Exchange. Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.
regulatoryHaltReason	1	Enum	Enum RegulatoryHaltReason . Regulatory reasons for trading halts and pauses. Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.

3.2 DefineSymbol

Symbol definition containing instrument details and trading parameters for a security. A DefineSymbol will be followed by a SymbolStatus message establishing the initial trading state for the symbol. Clients must treat a symbol as operationally HALTED while awaiting the accompanying SymbolStatus.

Field Name	Offset	Size	Type	Notes		
messageType	0	1	Byte	's'/0x73/115		
transactTime	1	8	TimeStamp			
symbolId	9	2	Short			
symbol	11	8	Str(8)	Security identifier root represented in CMS format.		
suffix	19	8	Str(8)	Security identifier suffix represented in CMS format.		
matchingEngineId	27	1	Byte	The matching engine to which this symbol is assigned for the trading session.		
defineSymbolBitFields	28	1	Bit Field Name	Offset	Size	Notes
			isTest	0	1	Test symbol flag.
			reserved	1	7	Reserved for future use.
lotSize	29	4	Int	The number of shares constituting one round lot.		
listingMarket	33	1	Enum	Enum MarketParticipantId . Market participant identifier of the listing exchange. Set to TXSE when TXSE is the listing market for this symbol; NOT_AVAILABLE (0x00) otherwise.		

3.3 SymbolStatus

Symbol-specific trading status representing the current state and restrictions for an individual security. A SymbolStatus is sent for every symbol, including those that open in the TRADING state. Clients must treat a symbol as operationally HALTED while awaiting the first SymbolStatus. Subsequent SymbolStatus messages are sent whenever the trading state or short-sale restriction changes.

Field Name	Offset	Size	Type	Notes
messageType	0	1	Byte	'y'/0x79/121
presenceBits	1	1	Byte	See Optional Fields
transactTime	2	8	TimeStamp	
symbolId	10	2	Short	
tradingState	12	1	Enum	Enum SymbolTradingState . Symbol-specific trading state indicating current availability.
shortSaleRestrictionState	13	1	Enum	Enum ShortSaleRestrictionState . Short sale restriction state under SEC Rule 201.

3.3.1 SymbolStatus Presence Bits

These presence bits are as follows. Any added optional fields must be in their presence bit order [See Optional Fields](#)

Bit Name	Set Mask
hasOperationalHaltReason	0x01
hasRegulatoryHaltReason	0x02
reserved	0xFC

3.3.2 SymbolStatus Optional Fields

The optional fields begin directly after the end of the field above. If a bit is set in the presence bits, the field must be present, and in the order listed in the presence bits. [See Optional Fields](#)

Field Name	Size	Type	Notes
operationalHaltReason	1	Enum	Enum OperationalHaltReason . Operational and technical reasons for trading halts initiated by the Exchange. Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.
regulatoryHaltReason	1	Enum	Enum RegulatoryHaltReason . Regulatory reasons for trading halts and pauses. Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.

3.4 Trade

An execution report for a single fill event in the order book. Quantity is the number of shares in this execution. An aggressive order that sweeps multiple resting orders produces one Trade per resting order filled.

Field Name	Offset	Size	Type	Notes
messageType	0	1	Byte	'x'/0x78/120
transactTime	1	8	TimeStamp	
symbolId	9	2	Short	
price	11	8	Long	
qty	19	4	Int	
tradeThroughExemptionReason	23	1	Enum	TradeThroughExemptionReason : NONE (0), INTERMARKET_SWEEP (1). Sale condition byte 2: Whether this trade is exempt from Reg NMS Rule 611 trade-through protection (CTA/UTP sale condition category/level 2). Additional values may be introduced in future Exchange releases without changing the message layout. Members should be prepared to receive values not listed here, and should treat any such value as a generic, unspecified member of this set rather than rejecting the message or ending the session.

3.5 TopOfBook

Current best bid and offer for a symbol. The BBO is the most aggressive price at which cumulative displayed liquidity reaches a round lot. Quantity is the total displayed shares at the BBO price combined with any displayed shares at more aggressive prices on that side. Price and quantity are zero when cumulative displayed liquidity does not reach a round lot. A TopOfBook is published whenever a depth change occurs at or inside the current BBO on the affected side. A TopOfBook may be published with values identical to the previous one if

the net result for the BBO happens to be unchanged after a relevant quantity change. When an execution causes both Trade and TopOfBook messages, all Trade messages from that execution event are delivered before the TopOfBook that reflects the resulting depth change.

Field Name	Offset	Size	Type	Notes
messageType	0	1	Byte	'q'/0x71/113
transactTime	1	8	TimeStamp	
symbolId	9	2	Short	
bidPrice	11	8	Long	Best bid price. Zero when cumulative displayed bid liquidity does not reach a round lot.
bidQty	19	4	Int	Total cumulative displayed quantity at the BBO bid price and all more aggressive bid prices. Zero when cumulative displayed bid liquidity does not reach a round lot. Capped at the maximum value of a 32-bit signed integer ; when the cumulative displayed quantity reaches this cap, the field indicates at least that many shares rather than an exact count.
askPrice	23	8	Long	Best ask price. Zero when cumulative displayed ask liquidity does not reach a round lot.
askQty	31	4	Int	Total cumulative displayed quantity at the BBO ask price and all more aggressive ask prices. Zero when cumulative displayed ask liquidity does not reach a round lot. Capped at the maximum value of a 32-bit signed integer ; when the cumulative displayed quantity reaches this cap, the field indicates at least that many shares rather than an exact count.

3.6 AuctionPreamble

Published every 5 seconds during the Accumulation Period to indicate projected auction size. Reports the matched volume and excess side that would result if the auction crossed at that moment, considering only Auction-Only orders; the Continuous Book is excluded. Emitted for OPENING and CLOSING auction types only. Not emitted for HALT or IPO auctions, which proceed directly to the Band Window phase at onset (see AuctionBandWindow).

Field Name	Offset	Size	Type	Notes
messageType	0	1	Byte	'u'/0x75/117
transactTime	1	8	TimeStamp	
symbolId	9	2	Short	
auctionType	11	1	Enum	Enum AuctionType . Type of auction event.
auctionStart	12	8	TimeStamp	Scheduled cross time for this auction, expressed as nanoseconds since the Unix epoch (UTC). For Opening and Closing auctions, reflects the Exchange's published session schedule. For Halt and IPO auctions, reflects the cross time announced for this specific event.
matchedShares	20	4	Int	Projected matched volume among Auction-Only orders.
excessAuctionSide	24	1	Enum	Enum AuctionExcessSide . Side with surplus Auction-Only volume.
reserved	25	8	Long	Reserved for future use. Set to 0 by the Exchange; receivers must ignore.

3.7 AuctionBandWindow

Published every 5 seconds during the Band Window to report Auction-Only volume positioned at each Participation Band boundary. The Participation Band is a volatility-based price range; its derivation methodology

is described in the Participation Bands specification. Auction-Only orders entered during the Band Window that are priced outside the band have their price adjusted to the nearest boundary. The Continuous Book is excluded. Volume counting at boundaries: a limit order is counted at each boundary price at which it would be eligible to execute; a market order is counted at both boundaries. Example: a limit buy at \$10.05 counts at the lower boundary if lower = \$10.05, but not at the upper boundary if upper = \$10.10.

Field Name	Offset	Size	Type	Notes
messageType	0	1	Byte	'v'/0x76/118
transactTime	1	8	TimeStamp	
symbolId	9	2	Short	
auctionType	11	1	Enum	Enum AuctionType . Type of auction event.
auctionStart	12	8	TimeStamp	Scheduled cross time for this cycle, expressed as nanoseconds since the Unix epoch (UTC). Updated to the new scheduled cross time on each extension.
extensionCycleCount	20	1	Byte	Extension cycle number. 0 on the first Band Window cycle. Incremented by 1 on each extension triggered by a projected clearing price outside the Participation Band. Always 0 for OPENING and CLOSING auctions, which do not support Band Window extensions.
lowerParticipationBand	21	8	Long	Price of the lower boundary of the Participation Band. Auction-Only sell orders entered during the Band Window and priced below this value have their price adjusted to this price.
buySharesAtLower	29	4	Int	Total shares of Auction-Only buy interest eligible to execute if the auction cleared at the lower Participation Band boundary.
sellSharesAtLower	33	4	Int	Total shares of Auction-Only sell interest eligible to execute if the auction cleared at the lower Participation Band boundary.
upperParticipationBand	37	8	Long	Price of the upper boundary of the Participation Band. Auction-Only buy orders entered during the Band Window and priced above this value have their price adjusted to this price.
buySharesAtUpper	45	4	Int	Total shares of Auction-Only buy interest eligible to execute if the auction cleared at the upper Participation Band boundary.
sellSharesAtUpper	49	4	Int	Total shares of Auction-Only sell interest eligible to execute if the auction cleared at the upper Participation Band boundary.
reserved	53	8	Long	Reserved for future use. Set to 0 by the Exchange; receivers must ignore.

3.8 AuctionPrint

Summary of an auction cross. Emitted before any per-order auction execution messages for the same cross. Universal lifecycle terminator — emitted for every auction. Encoding: (a) `matchedShares > 0` with a valid `auctionPrice` indicates a normal cross; (b) `matchedShares = 0` with a valid `auctionPrice` indicates a zero-share completion (e.g., halt-into-close default or demand failure); (c) `matchedShares = 0` with `auctionPrice = CANCELED_SENTINEL` indicates a canceled auction. Where the feed publishes per-order auction executions, those executions share this message's `execId`.

Field Name	Offset	Size	Type	Notes
messageType	0	1	Byte	'n'/0x6e/110
transactTime	1	8	TimeStamp	
symbolId	9	2	Short	
auctionType	11	1	Enum	Enum AuctionType . Type of auction event.
auctionPrice	12	8	Long	Auction clearing price.
matchedShares	20	4	Int	Total shares executed in the cross, across all order types including both Auction-Only and Continuous Book orders. Note: the matchedShares field in AuctionPreamble reports projected Auction-Only volume only; the two fields are not directly comparable.
execId	24	8	Long	Cross execution identifier. Where the feed publishes per-order auction executions, those executions carry this same execId for the cross. Unique per symbol per trading day; not guaranteed unique across symbols or across trading days.